

An Investigation of Sonar Detection on a Micro controller in a Real Time Application

Introduction

For this project I want to use an Arduino connected to three sonar sensors to read the detection signals of the sonar signals with the Arduino using solely interrupts. I want to consistently and accurately detect signals from the sonar sensors and investigate if distance can be measured with the sonar sensors using this method.

Is it possible to use interrupts to detect sonar readings from multiple sonars on an Arduino in a real time scenario?

Background Literature

In [1], [5], and [6] general info about Arduino is given. While [1] is a general introduction document to using Arduino in general, [5] and [6] are general resources and documentation about the Arduino UNO R3 specifically. However [1] is still useful for introduction as it complements [5] being an introduction and general source for the Arduino UNO R3 which then allows the contents of [6] to be navigated more easily when needing to find information for troubleshooting more in depth knowledge about the UNO R3.

The information in [2] as a source of general information for the HR-SR04 Ultrasonic sensor will be needed to understand how the sensor works with the Arduino using the information primarily from [5] and [6].

In [3] and [4] the hardware and circuit layouts are given for the UNO R3 which will probably needed to some degree for getting the UNO R3 to physically interact with the ultra sonic sensors and other possibly needed hardware that may need to be connected.

The implementation examples of using the Arduino UNO R3 with the HR-SR04 ultra sensor as shown in [7] and [8] together show how to implement both components of my project. While both [7] and [8] both have an Arduino working with a ultrasonic sensor, [7] has more emphasis on having the sonar and Arduino work together that will be needed for initial construction and implementation. [8] has more scope when it comes to working in a real time scenario with the Arduino where multitasking is needed which will be needed to take data from multiple sonars at the same time.

[9] will be important for most resources for code and general information when troubleshooting specifically as the git has code, examples, and specs on hardware and software. After being done with

the introductory information in [1], [5], [6], and [2], [9] will be my most valuable source of information for finding information beyond what is general to complete the project.

[10] will be important as the Arduino IDE is the main reason why using an Arduino is a good choice for the project. Since much of the implementation of this project will be in the firmware to make the Arduino and ultrasonic sensors work with each other [10] in particular will be important for troubleshooting the Arduino IDE and getting it to work.

Methodology

Starting Sunday April 26th, I will start setting up the Arduino and finish setup by that Sunday. Over the next week a plan on gaining basic familiarity and figuring out the general behavior of the Arduino and sonar sensors. By Friday May 1st I will be done with the physical implementation of the project. I will run various tests after physical and firmware setup of the Arduino and sonar sensors is finished to observe and record the behavior and outputs of the sonar sensors and see if I can get the Arduino to use interrupts to accurately detect signals from the sonar sensors simultaneously. The rates of false positives and false negatives to true positives and true negatives is going to be an incredibly important criteria monitor as this will show how accurate the Arduino is at detecting the sonar signals. The DoE is an implementation as I will be using a physical Arduino and sonar sensors.

Sources Cited

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